

Scalability & Future Plan

Date	05 July 2026
Team ID	[Enter Team ID]
Project Name	HDI Predictor
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Lightweight Flask threaded development server	Cannot handle high concurrent traffic loads.	Medium
2	Tabular CSV file backend storage	Poor write performance and scaling for high concurrent read/write query flows.	Medium
3	CPU-based local prediction routing	Inference bottleneck under heavy request concurrency.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Light Flask thread	Deploy backend with Gunicorn WSGI service behind Nginx reverse proxy.
2	Data Storage	Tabular CSV	Migrate country developmental indicators to PostgreSQL database.
3	Performance	Sub-50ms inference	Cache repeated coordinate queries with Redis key-value store.
4	Security	Local form checks	Implement HTTPS, secure SSL headers, and API rate limiting.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Auth & Scenario Saving	Q4 2026	Users can save developmental trajectories over time.

Phase 3	Multi-year Trend Forecasting	Q1 2027	Predictive forecasting of country HDI scores across future years.
Phase 4	Automated Data Sync	Q3 2027	Daily data ingestion sync pipeline with UNDP public API endpoints.